Presentation

INVENTORY DEMAND FORECASTING

- Predict demand
- Plan stock
- Reduce waste

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Introduction

Inventory Demand Forecasting is the process of predicting the future demand for products by analysing historical sales data and identifying patterns and trends.

This project uses data analysis and machine learning techniques to estimate future product demand. By studying previous sales records, the system can help businesses make better purchasing decisions and improve customer satisfaction.

The main aim of this project is to make inventory management smarter, more accurate and more efficient through data-driven demand prediction

Problem Statement

Many businesses face difficulties in managing inventory because they cannot accurately predict future product demand.

Major Problems :

- ***Overstocking increases storage costs.***
- ***Understocking leads to lost sales.***
- ***Manual demand estimation can be inaccurate.***
- ***Changing customer preferences affect demand.***
- ***Poor inventory planning wastes resources.***

Project Objectives

- ***Predict future demand using historical sales data.***
- ***Reduce overstocking and understocking.***
- ***Improve inventory planning and management.***
- ***Identify sales trends and seasonal patterns.***
- ***Support data-driven business decisions.***
- ***Improve product availability and customer satisfaction.***

Scope of the project

This project can be used by different types of businesses to manage inventory efficiently.

Applications

- ***Retail stores and supermarkets.***
- ***E-commerce platforms.***
- ***Warehouses and distribution centres.***
- ***Manufacturing industries.***
- ***Pharmaceutical and grocery businesses.***

Working Process

- *The Inventory Demand Forecasting system follows these steps:*
- *Data Collection: Collect historical sales and inventory data.*
- *Data Preprocessing: Handle missing values, duplicates, and incorrect data.*
- *Exploratory Data Analysis: Identify trends and seasonal patterns.*
- *Feature Engineering: Create useful features such as month, day, and previous sales.*
- *Model Training: Apply forecasting techniques to historical data.*
- *Model Evaluation: Measure forecasting accuracy.*
- *Demand Prediction: Estimate future product demand.*
- *Inventory Planning: Use predictions to make stock decisions.*

Data Collection

Data collection is the first step in building a demand forecasting system.

Important Data Fields :

- *Product ID*
- *Product name or category*
- *Date of sale*
- *Quantity sold*
- *Product price*
- *Available inventory*

Data Preprocessing

Data preprocessing prepares raw data for analysis and forecasting.

Main Steps:

- *Handle missing values.*
- *Remove duplicate records.*
- *Correct inconsistent data.*
- *Convert dates into the correct format.*
- *Identify unusual values.*
- *Organise sales records by date and product.*
- *Split data into training and testing sets.*

Exploratory Data Analysis (EDA)

Exploratory Data Analysis helps us understand sales data before applying forecasting models.

Main Activities:

- *Analyse daily and monthly sales.*
- *Identify high-demand products.*
- *Detect seasonal trends.*
- *Compare product categories.*
- *Identify unusual changes in demand.*
- *Create charts and graphs.*

Demand Forecasting Techniques

Different techniques can be used to predict future product demand

1. Moving Average

Calculates the average demand over a selected period.

2. Linear Regression

Uses relationships between variables to estimate demand.

3. Random Forest

Combines multiple decision trees to make predictions.

Feature Engineering and Model Training

Feature engineering creates useful variables that help a forecasting model identify patterns.

Examples of Features:

- *Day of the week*
- *Month and year*
- *Previous sales*
- *Average sales over recent periods*
- *Product price*

Model Training Process:

- *Select relevant features.*
- *Divide the data chronologically into training and testing sets.*
- *Train the selected model on historical data.*
- *Generate predictions for the testing period.*
- *Compare predictions with actual demand.*

Model Evaluation

Model evaluation measures how accurately a forecasting model predicts demand.

Common Evaluation Metrics:

1. MAE – Mean Absolute Error

Measures the average absolute difference between actual and predicted demand.

2. MSE – Mean Squared Error

Calculates the average squared prediction error.

3. RMSE – Root Mean Squared Error

Measures prediction error in the same units as demand.

4. MAPE – Mean Absolute Percentage Error

Measures average percentage error when actual values are non-zero and suitable for percentage-based evaluation

Benefits, Limitations, and Future Scope

Benefits

- *Helps reduce overstocking.*
- *Reduces the risk of stock shortages.*
- *Improves inventory planning.*
- *Supports better purchasing decisions.*
- *Helps control storage costs.*

Limitations

- Depends on historical data quality.
- Unexpected events can affect demand.
- New products may lack sufficient data.
- Predictions cannot guarantee actual demand.

Future Scope

- Real-time inventory tracking.
- Automated low-stock alerts.
- Interactive forecasting dashboard.
- Integration with inventory databases.

Conclusion

Inventory Demand Forecasting helps businesses predict future product demand by analysing historical sales data.

It supports better inventory planning, reduces unnecessary costs, minimises the risk of stock shortages, and improves product availability.

By combining data preprocessing, data analysis, forecasting techniques, and model evaluation, businesses can make more informed inventory decisions.

Thank You